



Panduan Uji Coba Hari ke-6 – Midtrans

QRIS Integration (POS-10A & POS-11)

Sprint 2 | Status: READY TO TEST

File ini berisi panduan lengkap untuk setup, testing, dan debugging integrasi Midtrans QRIS.



File yang Dibuat Hari Ini

File	Type	Keterangan
<code>config/midtrans.php</code>	Config	Konfigurasi Midtrans (server key, mode, dll)
<code>.env</code>	Environment	Variabel env termasuk Midtrans keys
<code>app/Models/DetailPembayaranNonTunai.php</code>	Model	Eloquent model tabel <code>detail_pembayaran_non_tunai</code>
<code>app/Models/Transaksi.php</code>	Model (Update)	Ditambahkan relasi <code>hasOne</code> ke <code>DetailPembayaranNonTunai</code>
<code>app/Services/MidtransService.php</code>	Service	<code>generateQris()</code> + <code>verifySignatureKey()</code>
<code>app/Http/Requests/V1/GenerateQrisRequest.php</code>	Request	Validasi input endpoint QRIS
<code>app/Http/Resources/DetailPembayaranResource.php</code>	Resource	JSON transformer untuk response payment
<code>app/Http/Controllers/Api/V1/PaymentController.php</code>	Controller	3 endpoint payment
<code>routes/api.php</code>	Routes (Update)	3 route baru ditambahkan



STEP 1: Setup Midtrans Sandbox

1.1 Dapatkan Kredensial Sandbox

1. Buka <https://dashboard.sandbox.midtrans.com>

2. Register/Login dengan akun Midtrans Anda
3. Navigasi ke: **Settings** → **Access Keys**
4. Salin **Server Key** (format: `SB-Mid-server-xxxxxxxx...`)
5. Salin **Client Key** (format: `SB-Mid-client-xxxxxxxx...`)

1.2 Update `.env`

Buka `pos-event-backend/.env` dan ganti nilai placeholder:

```
MIDTRANS_SERVER_KEY=SB-Mid-server-XXXXXXXXXXXXXXXXXXXXXXXXXXXX
MIDTRANS_CLIENT_KEY=SB-Mid-client-XXXXXXXXXXXXXXXXXXXXXXXXXXXX
MIDTRANS_IS_PRODUCTION=false
MIDTRANS_IS_SANITIZED=true
MIDTRANS_IS_3DS=true
MIDTRANS_QRIS_ACQUIRER=gopay
MIDTRANS_QRIS_EXPIRY_MINUTES=15
```

1.3 Konfigurasi Webhook URL di Dashboard Midtrans

Navigasi ke: **Settings** → **Configuration** → **Payment Notification URL**

Isi dengan URL webhook endpoint:

- **Development (ngrok):** `https://xxxx.ngrok.io/api/v1/payment/webhook`
- **Production:** `https://your-domain.com/api/v1/payment/webhook`

STEP 2: Setup ngrok untuk Testing Webhook Lokal

Midtrans membutuhkan URL publik untuk mengirim webhook. Gunakan ngrok:

```
# Install ngrok dari https://ngrok.com/download
# Jalankan server Laravel terlebih dahulu
php artisan serve --port=8000

# Di terminal lain, jalankan ngrok
ngrok http 8000
```

Salin URL HTTPS dari output ngrok (contoh: `https://abc123.ngrok.io`) dan set sebagai Notification URL di dashboard Midtrans.



STEP 3: Testing Endpoint via Postman / cURL

Pre-requisite: Dapatkan Bearer Token Kasir

```
curl -X POST http://localhost:8000/api/v1/auth/login/kasir \
-H "Content-Type: application/json" \
-d '{
  "username": "kasir01"
}'
```

Simpan token dari response: `"token": "1|xxxxx..."` untuk digunakan di request berikutnya.

3.1 – Buat Draft Transaksi (Pre-requisite untuk POS-10A)

Sebelum generate QRIS, buat draft transaksi dulu:

```
curl -X POST http://localhost:8000/api/v1/checkout/draft \
-H "Content-Type: application/json" \
-H "Authorization: Bearer TOKEN_KASIR_DI_SINI" \
-d '{
  "id_shift": "UUID_SHIFT_AKTIF",
  "id_cabang": "UUID_CABANG",
  "id_sales": "UUID_SALES_MODE",
  "id_metode": "UUID_METODE_QRIS_DYNAMIC",
  "items": [
    {
      "id_produk": "UUID_MENU",
      "harga_produk": 25000,
      "quantity": 2
    }
  ]
}'
```

Simpan `id_transaksi` dari response untuk digunakan di step berikutnya.

3.2 – POS-10A: Generate QR Code QRIS Dinamis

Endpoint: `POST /api/v1/payment/qr`

Auth: Bearer Token(Sanctum)

```
curl -X POST http://localhost:8000/api/v1/payment/qr \
-H "Content-Type: application/json" \
-H "Authorization: Bearer TOKEN_KASIR_DI_SINI" \
-d '{
  "id_transaksi": "UUID_TRANSAKSI_DRAFT_DI_SINI"
}'
```

Contoh Response Sukses (HTTP 200):

```
{
  "success": true,
  "message": "QR Code QRIS berhasil di-generate. Silakan scan menggunakan",
  "data": {
    "id_detail_bayar": "550e8400-e29b-41d4-a716-446655440000",
    "id_transaksi": "UUID_TRANSAKSI",
    "payment_gateway_id": "77af9e71-4ce1-4eb0-bcdc-3f77ea1d0c9c",
    "reference_number": null,
    "qr_string_data": "https://api.sandbox.midtrans.com/v2/qr/77af9e71-",
    "va_number": null,
    "status_api": "PENDING",
    "waktu_kedaluwarsa": "2026-07-22T00:00:00+07:00",
    "transaksi": null,
    "created_at": "2026-07-21T23:45:00+07:00",
    "updated_at": "2026-07-21T23:45:00+07:00"
  }
}
```

3.3 – Polling Status (Cek apakah sudah dibayar)

Endpoint: `GET /api/v1/payment/status/{id_transaksi}`

Auth: Bearer Token(Sanctum)

```
curl -X GET http://localhost:8000/api/v1/payment/status/UUID_TRANSAKSI_DI
-H "Authorization: Bearer TOKEN_KASIR_DI_SINI"
```

Contoh Response (HTTP 200):

```
{
  "success": true,
  "message": "Data status transaksi berhasil diambil.",
  "data": {
    "id_transaksi": "UUID_TRANSAKSI",
    "status": "Pending",
    "total": 55000.0,
    "pembayaran": {
      "payment_gateway_id": "77af9e71-4ce1-4eb0-bcdc-3f77ea1d0c9c",
      "status_api": "PENDING",
      "waktu_kedaluwarsa": "2026-07-22T00:00:00+07:00",
      "sis_a_detik": 842,
      "sudah_dibayar": false,
      "sudah_expired": false
    }
  }
}
```

3.4 – POS-11: Simulasi Webhook Midtrans (Manual Test)

Endpoint: POST /api/v1/payment/webhook

Auth: TIDAK ADA (Public endpoint)

Cara Membuat signature_key Manual

Formula:

SHA-512 (order_id + status_code + gross_amount + server_key)

Contoh (PowerShell):

```
$orderId      = "UUID_TRANSAKSI_ANDA"
$statusCode   = "200"
$grossAmount  = "55000.00"
$serverKey    = "SB-Mid-server-XXXXXXXXXXXXXXXXXXXXXXXXXXXX"

$raw = $orderId + $statusCode + $grossAmount + $serverKey
$sha512 = [System.Security.Cryptography.SHA512]::Create()
$bytes  = [System.Text.Encoding]::UTF8.GetBytes($raw)
$hash   = $sha512.ComputeHash($bytes)
$signatureKey = [BitConverter]::ToString($hash).Replace("-", "").ToLower()
Write-Output $signatureKey
```

Contoh (Linux/Mac bash):

```
ORDER_ID="UUID_TRANSAKSI_ANDA"  
STATUS_CODE="200"  
GROSS_AMOUNT="55000.00"  
SERVER_KEY="SB-Mid-server-XXXXXXXXXXXXXXXXXXXXXXXXXXXX"
```

```
echo -n "${ORDER_ID}${STATUS_CODE}${GROSS_AMOUNT}${SERVER_KEY}" | sha512s
```

Payload Webhook Simulasi – Pembayaran BERHASIL (settlement)

```
curl -X POST http://localhost:8000/api/v1/payment/webhook \  
-H "Content-Type: application/json" \  
-d '{  
  "transaction_time": "2026-07-22 00:00:00",  
  "transaction_status": "settlement",  
  "transaction_id": "77af9e71-4ce1-4eb0-bcdc-3f77eald0c9c",  
  "status_message": "midtrans payment notification",  
  "status_code": "200",  
  "signature_key": "SIGNATURE_SHA512_YANG_SUDAH_DIHITUNG",  
  "settlement_time": "2026-07-22 00:05:00",  
  "payment_type": "qris",  
  "order_id": "UUID_TRANSAKSI_ANDA",  
  "merchant_id": "G000000000",  
  "gross_amount": "55000.00",  
  "fraud_status": "accept",  
  "currency": "IDR",  
  "acquirer": "gopay"  
}'
```

Expected Response (HTTP 200):

```
{  
  "status": "success",  
  "message": "Webhook processed successfully."  
}
```

Payload Webhook Simulasi – QR EXPIRED

```
curl -X POST http://localhost:8000/api/v1/payment/webhook \
-H "Content-Type: application/json" \
-d '{
  "transaction_time": "2026-07-22 00:00:00",
  "transaction_status": "expire",
  "transaction_id": "77af9e71-4ce1-4eb0-bcdc-3f77ea1d0c9c",
  "status_code": "407",
  "signature_key": "SIGNATURE_SHA512_UNTUK_STATUS_407",
  "payment_type": "qris",
  "order_id": "UUID_TRANSAKSI_ANDA",
  "gross_amount": "55000.00",
  "currency": "IDR"
}'
```

Payload Webhook Simulasi – DITOLAK/DENIED

```
curl -X POST http://localhost:8000/api/v1/payment/webhook \
-H "Content-Type: application/json" \
-d '{
  "transaction_time": "2026-07-22 00:00:00",
  "transaction_status": "deny",
  "transaction_id": "77af9e71-4ce1-4eb0-bcdc-3f77ea1d0c9c",
  "status_code": "202",
  "signature_key": "SIGNATURE_SHA512_UNTUK_STATUS_202",
  "payment_type": "qris",
  "order_id": "UUID_TRANSAKSI_ANDA",
  "gross_amount": "55000.00",
  "fraud_status": "deny",
  "currency": "IDR"
}'
```

Test Keamanan – Invalid Signature (harus HTTP 403)

```
curl -X POST http://localhost:8000/api/v1/payment/webhook \
-H "Content-Type: application/json" \
-d '{
  "transaction_status": "settlement",
  "transaction_id": "fake-id",
  "status_code": "200",
  "signature_key": "invalid_signature_should_fail",
  "order_id": "UUID_TRANSAKSI_ANDA",
}
```

```
"gross_amount": "55000.00"
}'
```

Expected Response (HTTP 403):

```
{
  "success": false,
  "message": "Invalid Signature Key."
}
```



Alur Status Transaksi Lengkap

Draft → [generate QRIS] → Pending → [webhook settlement] → Success
→ [webhook expire] → Cancelled
→ [webhook deny] → Cancelled

Mapping Status Detail

Midtrans transaction_status	detail_pembayaran_non_tunai.stat us_api	transaksi.sta tus
settlement	SETTLEMENT	Success
capture + fraud=accept	SETTLEMENT	Success
capture + fraud≠accept	DENIED	Cancelled
expire	EXPIRED	Cancelled
deny	DENIED	Cancelled
cancel	DENIED	Cancelled
pending	tidak berubah	tidak berubah



Troubleshooting

QR String Kosong

- Pastikan `MIDTRANS_QRIS_ACQUIRER=gopay` di `.env`
- Pastikan akun Midtrans Sandbox sudah mengaktifkan QRIS/GoPay
- Cek log di `storage/logs/laravel.log`

Webhook Tidak Diterima

- Pastikan ngrok berjalan dan URL sudah diset di Dashboard Midtrans
- Cek apakah URL dapat diakses dari luar: `curl`

`https://xxxx.ngrok.io/api/v1/payment/webhook`

Signature Invalid

- Pastikan `gross_amount` dari webhook cocok persis (termasuk desimal, contoh: `"55000.00"` bukan `"55000"`)
- Pastikan `MIDTRANS_SERVER_KEY` di `.env` sesuai dengan yang ada di Dashboard Midtrans